

Feb 20, 2026

Timur / Jesus

Invited [Jesus Gonzalez Godoy](#) [Timur Khamidulin](#)

Attachments [📎 Timur / Jesus](#)

Meeting records [📄 Transcript](#)

Summary

Pipeline Status and Application Structure

The pipeline development was confirmed as complete and operational, accepting queries and delivering results, though some additional evaluation work is required. A significant discussion centered on converting the current interactive command-line application into a reusable framework by implementing it in a notebook format.

Debugging and Code Documentation Standards

The participants engaged in debugging errors found during script execution, particularly within the `monitor` class initialization and later during the vector database update. The importance of adding doc strings to all AI-generated code, especially in the `monitor` class, was emphasized for clearer documentation and usability.

Identifying and Resolving Data Errors

Debugging efforts localized a recurring error to the processing of PDF files in the GCP folder, identifying that one document was likely corrupted or built differently. The team concluded that the problematic PDF should be removed and replaced, as fixing the file corruption was outside the project scope.

Details

Decisions

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ALIGNED

- **Framework runnable notebook creation** The RAG framework implementation must also include a runnable notebook version, requiring the removal of the ``main`` entry point from the script.
- **Include doc strings for AI code** Doc strings must be included for every class generated by AI to improve code documentation and understanding.
- **Remove TXT/MD file indexing code** Extraneous code blocks related to indexing TXT and MD files must be removed from the pipeline, as only PDF files are intended for indexing.
- **Corrupted PDF file removal** The corrupted PDF file should be removed and replaced with another one because repairing it is outside the scope of the current project.
- **Code restructuring and cleanup tasks** Specific preparatory tasks for the next project check-in are assigned: cleaning the code, restructuring it to work as a framework instead of an interactive application, analyzing short answers using prompt tricks, making rules functional, and finalizing the evaluation metrics.

More details:

- **Initial Check-in and Project Status:** Timur Khamidulin and Jesus Gonzalez Godoy confirmed the pipeline development is complete and operational, accepting queries and delivering results. Timur Khamidulin noted they still needed to complete some additional evaluation work ([00:00:00](#)).
- **Pipeline Demonstration and Code Review:** Timur Khamidulin shared their screen to demonstrate the working pipeline, which was loaded from the terminal ([00:01:27](#)). The pipeline showed the ability to retrieve answers from the source

PDF documents, such as answering "What is rock," though the initial response was noted by Jesus Gonzalez Godoy as being too short ([00:03:19](#)).

- **Review of Interactive Application Structure:** Jesus Gonzalez Godoy confirmed that the implementation functioned more like an interactive command-line application than a framework, due to the inclusion of a `main` function ([00:05:22](#)). Timur Khamidulin acknowledged that they could convert the application to a notebook format if preferred, and Jesus Gonzalez Godoy stated that the current method was acceptable but that removing the `main` function and utilizing a notebook was an alternative approach ([00:07:33](#)).
- **Identifying an Error in the Monitor Class:** Upon running the script, a debugging process began to identify an error related to "compressing data". By commenting out subsequent blocks of code, they determined the error was located within the `monitor` class initialization ([00:07:33](#)) ([00:11:30](#)).
- **Conceptual Difference Between Framework and Application:** Jesus Gonzalez Godoy clarified the difference in purpose between the current command-line application, which uses a defined interactive loop and `main` function, and a framework, which is designed to be used in another script or notebook. Timur Khamidulin committed to implementing the framework in a notebook format by the following Monday ([00:13:13](#)).
- **Importance of Doc Strings for Documentation:** Timur Khamidulin confirmed that they used an AI assistant (Gemini) to generate the structure for the `monitor` class, and Jesus Gonzalez Godoy stressed the importance of including doc strings in every AI-generated class for easier documentation ([00:14:27](#)). Using the provided AI assistant (Cloud), they added doc strings to the `monitor` class ([00:16:16](#)).
- **Illustrating Doc String Utility:** They demonstrated the practical utility of doc strings by hovering the mouse over the `monitor` instance, showing the generated description and signature, which helps anyone using the tool understand its function, inputs, and outputs ([00:18:59](#)). Jesus Gonzalez Godoy also noted that the `monitor` class, which uses a `timeit` function to measure time, was overly complex due to the AI generation, suggesting that a Python decorator might be a better approach ([00:21:30](#)).
- **Discussion on Responsible AI Use in Coding:** They discussed the observation that AI often generates well-done but unnecessarily long or complex code

(00:23:01). Jesus Gonzalez Godoy encouraged the use of AI tools but emphasized the necessity of understanding every line of code the AI creates and providing clear, well-guided input to ensure good quality output (00:24:49).

- **Debugging the Vector Database Update:** They returned to debugging and located an error when the script attempted to update the vector database (00:34:04). Jesus Gonzalez Godoy suggested that the error might be occurring because the document was being indexed again, or potentially due to an issue with one of the PDF files (00:35:57).
- **Investigating the `load_pages` Function:** They started debugging the `update_database` method and determined that the error was occurring inside the `load_pages` function (00:37:47) (00:43:03). After examining the code, Jesus Gonzalez Godoy identified unnecessary complexity, such as the use of `sorted` in loops for folders and files, and a large block of extraneous code intended to index TXT and MD files, which was not the project's goal (00:47:21) (00:51:37).
- **Pinpointing the Error Location in the GCP Folder:** By adding print statements before processing each topic folder, they localized the error to the PDF files being processed within the GCP folder. Jesus Gonzalez Godoy suggested they might try processing the PDF files one by one to isolate the problematic document (00:55:35).
- **Troubleshooting PDF Errors:** The speakers collaboratively debugged an error related to processing PDF files, which initially involved using a temporary folder and testing individual files to isolate the problematic document (00:57:47). After identifying which PDF was causing issues, they traced the error message "found zero objects within object 7075" and determined that the specific PDF was likely corrupted or built differently than the others, leading to an error loading that file. It was concluded that because fixing the PDF corruption was outside the project scope, the problematic file should simply be removed and replaced (01:00:05) (01:04:10).
- **Using AI Tools for Debugging:** Jesus Gonzalez Godoy advised Timur Khamidulin on using AI assistants like Cloud or Gemini for debugging, suggesting that when using a tool without context (like Gemini), they should first provide the context of the code and the desired answer (01:02:29). The immediate debugging steps included adding a print statement for exceptions inside the error handling loop, which helped display the specific error message "error loading that PDF because cannot find group object in PDF" (01:00:05).

- **Assignments and Next Steps for Code Refinement:** Timur Khamidulin committed to reviewing the code and understanding each line, stating they would complete this by Monday ([01:04:10](#)). Jesus Gonzalez Godoy tasked them with cleaning the code, ensuring it functions as a framework rather than an interactive application, and making sure the evaluation metrics are completed. Future project tasks will involve experimenting with short answers, tricking the prompt, and making the rules added to the prompt function correctly ([01:05:27](#)).
- **Scheduling the Next Meeting:** The speakers agreed to meet again on Monday following a lecture that Timur Khamidulin was scheduled to attend. They discussed potential meeting times, including 11 a.m. or a later evening call (7 p.m. Jesus Gonzalez Godoy's time / 8 p.m. Timur Khamidulin's time), with Jesus Gonzalez Godoy planning to send an invite for either 11 a.m. or the evening slot ([01:06:35](#)).

Suggested next steps

- ☐ Timur Khamidulin will create the notebook by Monday.
- ☐ Timur Khamidulin will include doc strings for every class, especially for the vector database, by Monday.
- ☐ Timur Khamidulin will go through and clean the code to understand each line, using cloud, and make it work as a framework instead of an interactive application by Monday.
- ☐ Jesus Gonzalez Godoy will check Monday's schedule and send an invite for 11 AM or evening.
- ☐ Timur Khamidulin will finish the evaluation.

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